Tamper-Proofing with Self-Modifying Code

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Abstract

Classical computability theory tells us that self-modifying code (SMC) on a deterministic universal Turing machine can be simulated by non-SMC code on the same model. That abstraction, however, omits the external timing inputs, concurrency, and microarchitectural state that dominate practical execution on modern processors. We argue that once timing, ordering, and self-introspective effects are treated as observables, a *practically faithful* non-SMC reproduction of timed SMC becomes detectably expensive on commodity systems. We present a tamper-proofing model that combines introspective and polymorphic SMC, reliable clocks, and runtime timing predicates to bind integrity checks to execution behavior. We distinguish static and dynamic SMC generation, characterize the timing semantics needed to avoid catastrophic pipeline clears, and give x86-64 design primitives for checksum-driven self-patching. We also report timer measurements, performance comparisons, and performance-monitoring counter evidence showing that careful engineering—especially loop unrolling and cross-page modification—substantially reduces the overhead of SMC while preserving its tamper-detection value. The paper concludes with an efficiency analysis, a threat model, and deployment guidance for trusted code executing in untrusted environments. The threat model is formalised in Lean 4, relative to an explicit platform contract: the security theorem, the necessity of each assumption and which assumptions each adversary level satisfies or breaks are machine-checked, and the development is registered with the Palomar registry.

Keywords: x86, x86-64, assembly language, self-modifying code, software protection, tamper-proofing, integrity checking, runtime verification

1 Introduction

It is classical in computability theory that self-modifying code (SMC) on a deterministic universal Turing machine (UTM) can be simulated by non-SMC code on the same model; SMC does not increase computational power beyond Turing completeness, but rather changes representation and potentially efficiency [1, 2, 3]. That baseline, however, abstracts away *external inputs*. In practice, time sources such as Coordinated Universal Time (UTC) and processor time-stamp counters (TSCs), together with concurrent scheduling, introduce nondeterminism: even if a TSC were made deterministic, UTC would not be; in parallel settings, TSC readings are further complicated by cache and memory synchronization.

Once such inputs are admitted as observables, the equivalence of SMC to non-SMC becomes fragile in a tamper-proofing context. Functional behavior can still be simulated in principle, but preserving *timing*, *ordering*, and *self-introspective effects* in the presence of an adversary generally cannot be achieved without either reintroducing self-modification or incurring detectable distortions.

External inputs: what helps, what does not. Many machine and system identifiers—CPU ID strings, NIC MAC addresses, disk/RAM configurations, or operating-system fingerprints—are useful for binding a license to a host [4], but they solve fingerprinting more directly than tamper-proofing and are operationally brittle. System-activity signals such as mouse movement or event logs either feed entropy pools [5] or exhibit regularity that attackers can mimic. Cloud- or server-provided data can be incorporated as an external input, but replay and protocol attacks complicate correctness unless additional machinery is deployed; moreover, such channels do not protect code at the machine-architecture level against manual debugging, thread injection, or remote process-memory writes. The mechanism proposed here does not prevent these either: it detects persistent modification of the checksummed bytes and interference inside timed intervals, but not injected code that leaves the checksummed bytes intact, register tampering outside timed intervals, or a modification that is undone before the next covering check (Section 10). A well-engineered network time source can, however, be a reliable UTC input.

Practical stance. For integrity checks that bind behavior to time, there is effectively no *practically faithful* non-SMC equivalent that preserves both semantics and timing in the face of an adversary on contemporary systems. An attacker who controls the time source need not model the processor at all: replaying readings recorded from native runs makes any threshold predicate on those readings pass. Timing-based detection therefore requires a time source outside the adversary's control, for example `RDTSC`/`RDTSCP` read inside the protected code against a user-mode adversary, or an external source against a kernel-level adversary. An adversary who must run the protected code under emulation with an honest clock needs a deterministic, fully characterizable timing model, and modern commodity processors do not admit one. Execution timing is influenced by partially undocumented microarchitectural state—branch predictors, caches, coherence protocols, speculative execution, microcode—together with concurrent activity and operating-system scheduling. Reproducing these effects within tight timing tolerances would require a faithful simulation of both the processor and the operating system, including their dynamic state, and doing so at practical speed is infeasible.

Model. We propose a tamper-proofing model in which SMC is (i) *introspective*, in that it checksums its own code, and (ii) *polymorphic*, in that it rewrites itself while executing. The model uses algorithmic kernels whose control flow and instruction count are independent of the data to reduce unrelated timing leakage and may treat sensitive data as code (SMC-backed storage) to harden against data-only faults. A virtualized SMC wrapper can dynamically generate an inner SMC layer. Formally, we assume a timer $T(t)$ chosen per query t for reliability and a checksum function $C(c)$ over code or data c; the predicate $P(T, C)$ returns true for the unmodified protected code (when its timing lies within the calibrated bounds) and false for every modification that changes the checksum state or violates the time bound.

Relation to prior work. Tamper-proofing via watermarking is well studied [6]. Self-checksumming with SMC has also been explored [7], but typically with static checksum routines rather than checksum kernels that rewrite themselves while executing. Remote runtime-integrity schemes can rely on a verifier in the cloud [8]; our goal is to avoid such infrastructure and keep the assurance mechanism local to the untrusted host. This inverts the conventional "trustworthy computing" perspective, which usually assumes a trusted host running untrusted software. Our setting instead asks how trusted software can retain integrity on an untrusted host, which is the operationally harder direction.

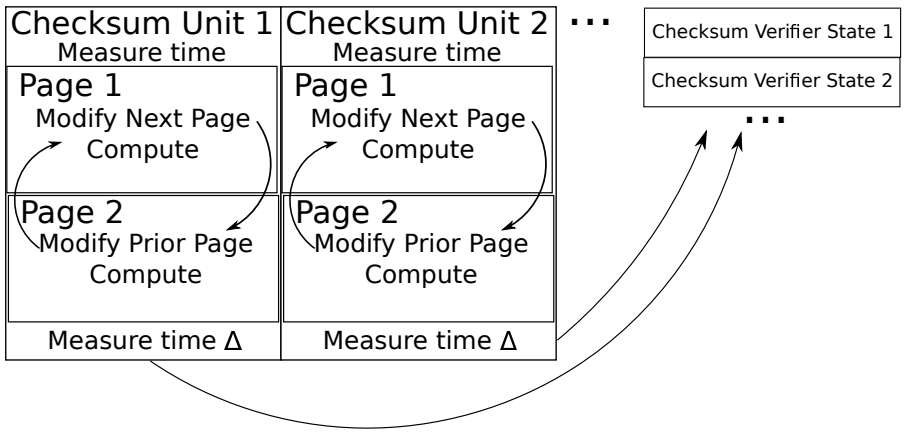


Figure 1: Design overview of checksum units and state verification.

Contributions. The paper makes four concrete contributions.

- It frames timed, introspective SMC as a practical tamper-proofing primitive whose non-SMC reproduction is detectably costly on modern systems.
- It develops a structured engineering model covering reliable clocks, timing semantics, static versus dynamic generation, and concrete x86-64 design primitives.
- It reports empirical timing and performance-counter results that explain when SMC is impractical and when it becomes efficient enough to deploy.
- It states explicit efficiency and security assumptions so that the scope and limitations of the approach are clear.

Figure 1 gives a high-level overview of the resulting design.

The remainder of the paper proceeds as follows. Section 2 evaluates practical time sources. Section 3 contrasts static and dynamic SMC generation. Section 4 formalizes the timing semantics relevant to detection. Sections 5 and 6 present static and dynamic design primitives. Section 7 reports measured performance. Section 8 interprets the cost model. Section 9 shows how the primitives compose into a deployment scheme. Section 10 states the threat model and assumptions, and Section 11 concludes.

2 Reliable clocks

The read TSC instructions `RDTSC` and `RDTSCP` are not perfectly reliable in isolation because modern processors expose hyper-threading, out-of-order execution, speculative execution, and cache effects that can perturb measurements [9]. Nevertheless, they can be made reliable enough statistically by repeated execution and by surrounding them with serialization instructions such as `CPUID` and `RDTSCP`. In many tamper-proofing contexts, a high-probability integrity decision is sufficient to continue execution.

Most contemporary systems also expose several other clocks. High Precision Event Timer (HPET), real-time clock (RTC) facilities, and operating-system monotonic counters provide alternative timing sources with different resolution and noise profiles. The specific x86 instructions discussed here are not unique in spirit: AVR32, SPARC V9, PowerPC, ARMv7, and ARMv8 all

provide related counter-style instructions, underscoring the near universality of hardware time measurement.

For x86 time-stamp counters, the instructions `RDTSC` and `RDTSCP` have the corresponding intrinsics `__rdtsc` and `__rdtscp`, which are readily usable from user mode. On many systems they offer the most fine-grained timing information available to ordinary software.

On Windows, several Application Programming Interface (API) calls expose timer state. UTC can be queried with `GetSystemTimePreciseAsFileTime`. The high-resolution performance counter is obtained with `QueryPerformanceCounter` and calibrated with `QueryPerformanceFrequency`. The interrupt timer is available through `QueryInterruptTimePrecise`. These three are practically in the same precision class. Coarser timers include the tick counter `GetTickCount64()` and multimedia timers via `timeBeginPeriod(1)`, `timeEndPeriod(1)`, and `timeGetTime()`, though they are not intended for precise microbenchmarking.

On Linux, `clock_gettime` provides access to several clocks. The system clock is obtained through `CLOCK_REALTIME`. A monotonic clock analogous to an interrupt timer is measured with `CLOCK_MONOTONIC`, and a boot-time clock including suspended time is available through `CLOCK_BOOTTIME`. UTC time can also be queried with `gettimeofday`, although `clock_gettime` is generally the cleaner interface in contemporary software. Standard C and C++ library abstractions wrap part of this functionality and may be easier to use when the implementation details do not matter.

3 Static versus dynamic generation of SMC

Static SMC is handwritten assembly code linked into the program at compile time. It is difficult to write and maintain because assemblers and C/C++ toolchains do not directly accommodate code that rewrites its own instruction bytes. Compiler support is also inconsistent: Microsoft Visual C++ supports inline assembly on x86 but not x86-64, whereas the GNU Compiler Collection allows inline assembly more broadly.

Dynamic SMC is generated at runtime, for example with an assembler library such as AsmJit [10]. Although such frameworks are typically used for just-in-time compilation, they also provide a convenient foundation for SMC that must be polymorphic or context sensitive. Features missing from conventional assemblers can be introduced programmatically, and the generated code can adapt to the operating-system version, processor family, address-space layout, or deployment policy at runtime.

Static code is normally loaded into read-execute (RX) pages. Since SMC needs write access, the binary or loader configuration must be adjusted accordingly. Under MSVC, this can be expressed at link time with `/SECTION:.text,RWE`. With other toolchains, such as GCC on Portable Executable (PE) or Executable and Linkable Format (ELF) targets, the required section permissions may need manual editing in the binary format metadata.

For dynamically generated SMC, the memory allocator must request pages with the appropriate permissions. On Windows, this is achieved with `VirtualAlloc(NULL, codeSize, MEM_RESERVE | MEM_COMMIT, PAGE_EXECUTE_READWRITE)`. On Linux, the corresponding mechanism is typically `mmap`. In both cases, deployment policies should be reviewed carefully if write-execute (WX/RWX) pages are restricted.

Dynamic SMC also makes address tracking more difficult for emulation tools. Because the code layout is generated at runtime, the addresses that a semantics- faithful non-SMC emulator must monitor are themselves contextual. This is one of the reasons dynamic generation is especially attractive for tamper-proofing.

Item	Intel x86-64	AMD64
Perf. Counter Global Control	0x38F	0xC0000301
Perf. Event Selector 0	0x186	0xC0010000
Perf. Counter 0	0xC1	0xC0010004
SMC Event Name / Unit Mask	`MACHINE_CLEARS.SMC`	`PIPELINE_RESTART_DUE_TO_ SELF_MODIFYING_CODE`
SMC Event Value	0xC3 / 0x4	0x21 / 0x0

Table 1: Intel x86-64 and AMD64 address and event reference values for representative supported microarchitectures.

4 Timing semantics of SMC

Efficient tamper-proofing with SMC requires careful attention to modern microarchitectural behavior, particularly instruction pipelines and memory hierarchies. Most processor optimizations of recent decades have focused on deep pipelines, speculative pre-processing, and aggressive caching. Na"ive SMC can invalidate that work and degrade execution to something resembling a much older processor running at a higher clock frequency. For a tamper-proofing scheme, such a penalty is unacceptable.

A practical workaround is to structure the checksum routine so that it does not modify the currently executing instruction stream. *Loop unrolling* is one useful technique: it increases the number of instructions executed between modification points and thus allows the pipeline to remain populated. When cache effects matter, unrolling can also distribute self-modifying regions across distinct pages, reducing the chance that the executing page is also the one being written.

Although modern processors ultimately remain von Neumann machines, their deep pre-execution machinery makes SMC expensive when it touches instructions that are already in flight. On x86, such cases can trigger a *machine clear*: the processor squashes in-flight instructions and restarts fetch/decode for the logical processor. This is a major source of slowdown. Similar effects can also appear in cross-modifying code (XMC) settings where one thread writes code later executed by another. Designs therefore have to account for these penalties and avoid them when possible.

The detection of SMC by the processor is itself approximate. It often uses the current page or a smaller region as the detection unit, which means false positives can occur when a modification is near, but not identical to, the bytes currently in flight. Consequently, both timing thresholds and code layout must be engineered with these approximations in mind.

Hardware performance counters help tune these designs. On x86, model-specific registers (MSRs) can configure event counters that are later read through `RDPMC` if control register 4 allows it. The configuration interface is vendor specific in its addresses but conceptually similar across Intel and AMD processors [11, 12]. Table 1 summarizes the key register and event identifiers used in our measurements.

Between Windows and Linux, the main difference is access control. Performance counter configuration is privileged. On Windows, practical experimentation may therefore require a signed kernel driver such as WinRing0 [13]. On Linux, `perf_event_open` can expose performance counters to user mode under policy, so a separate kernel module is not always necessary. Importantly, the counters were used only for *measurement and tuning*; deployment of the tamper-proofing mechanism itself does not require direct PMC access.

5 Static SMC tamper-proofing design primitives

To demonstrate the core idea, we begin with a simple static design. Efficient SMC depends heavily on the details of the instruction set architecture (ISA). Many ISAs contain opcode families

Instruction	Opcode	Opcode[3:5]	LAT	RCP
ADC	0x13	2	1	1
ADD	0x03	0	1	0.25
AND	0x23	4	1	0.25
CMP	0x3B	7	1	0.2 / 0.25
OR	0x0B	1	1	0.25
SBB	0x1B	3	1	1
SUB	0x2B	5	1	0.25
XOR	0x33	6	1	0.25

Table 2: x86 arithmetic instruction pattern used by the checksum kernel. Latency and reciprocal throughput are in cycles; ranges reflect microarchitectural variation [14].

ADC	Add with carry
SBB	Subtract with borrow
XOR	Exclusive OR
SHL	Shift left (logical, same as SAL)
LEA	Load effective address
SETcc	Set byte on condition
CMP	Compare operands
Jcc	Conditional jump
cc=S	Sign / negative (SF = 1)
cc=G	Greater than (signed, ZF = 0 and SF = OF)
cc=P	Parity (PF = 1)

Table 3: Reference for instructions used by the example checksum kernel. The parity flag uses the legacy low-8-bit parity computation.

whose encodings differ only in a few bits. On x86-64, this makes it possible to exchange arithmetic operations by toggling a small number of bits in place. In particular, several long-standing arithmetic instructions share a common format for a 64-bit register destination and a 64-bit register-or-memory source. Table 2 shows the useful subset.

By exploiting this encoding pattern and combining it with loop unrolling, we develop a checksum scheme. Here, a *checksum scheme* is a self-verification mechanism in which a program computes a lightweight integrity value over a region of its own code or data and uses that value at runtime to detect tampering or to select the next execution state. The checksum is not intended to be collision resistant in the cryptographic sense; the protection comes from coupling it to dynamic state and timing.

With multiple checksum functions, the program can maintain validation data for a set of reachable states defined by each checksum kernel's last modification. Reset points before or after initialization keep the state space manageable. Figure 2 shows a minimal SMC checksum kernel, while Figure 3 shows a C++ implementation that emulates the same behavior as faithfully as possible without using SMC.

One x86-64 feature is especially helpful here: instruction-pointer-relative addressing. Because RIP-relative addressing allows nearby code bytes to be reached without first materializing an absolute address in a register, the SMC kernel can patch itself with comparatively little scaffolding. Earlier 16-bit and 32-bit x86 variants do not provide this mechanism in the same form; there, the programmer typically needs an auxiliary register and a `CALL`/`POP` style idiom to recover the current address, which perturbs the stack and makes the checksum kernel less elegant.

The next static variant loop-unrolls the checksum body and modifies an opcode at some fixed

```
checksum:
ADC RBX, QWORD PTR [RSI] ; 48 13 1E
SETS DL ; 0F 98 C2
SETP AL ; 0F 9A C0
SHL AL, 2 ; C0 E0 02
OR DL, AL ; 0A D0
SHL DL, 3 ; C0 E2 03
XOR BYTE PTR [$-2-3-3-3-2-3], DL
LEA RSI, [RSI+8]
CMP RDI, RSI
JG checksum
```

Figure 2: A simple static SMC checksum variant.

```
const size_t offsetInFunc = 0xF;
// manually found based on MOV, LEA, XOR, CLC, ADC prefix
const PULONG64 pModOffset = (PULONG64)((ULONG64)((PBYTE)pTPCompOffset + offsetInFunc) & ~0x7ULL);
const unsigned char pModByteOffset = ((ULONG64)((PBYTE)pTPCompOffset + offsetInFunc) & 7) * 8 + 3;
// convert to bits, add 3 bits for the ADC/SBB/XOR/CMP offset
const unsigned __int64 preserve = ~(5ULL << pModByteOffset);
for (LPBYTE pEnd = (LPBYTE)pData + ulSize; (LPBYTE)pData < pEnd; pData++) {
    unsigned __int64 res;
    unsigned __int64 pDataVal = pData == pModOffset ?
        (*pData & preserve) |
        ((unsigned __int64)op << pModByteOffset) : *pData;
    switch (op) {
    case 0: res = sum += pDataVal; break;
    case 1: res = sum -= pDataVal; break;
    case 4: res = sum ^= pDataVal; break;
    case 5: res = sum - pDataVal; break;
    default: res = 0;
    }
    op ^= (!paritybyte((unsigned char)res) << 2) |
        ((res & 0x8000000000000000ULL) != 0);
}
```

Figure 3: Minimal C++ code to emulate the SMC checksum computation precisely. The function arguments are the start address `pData`, the protected length `ulSize` (a multiple of 8 bytes) and the result address `pResult`; `pTPCompOffset` is the address of the body of the kernel `TPComp` and is computed before the loop (not shown), `sum` is stored to `*pResult`, and `op` is the persistent opcode state. The emulation is exact for an 8-byte-aligned region when `op` equals the opcode state of the kernel at entry.

offset from each unrolled position. Unrolling has two advantages. First, it improves timing behavior: with a suitable offset and sufficient depth, pipeline clears can be reduced dramatically and may disappear altogether. Second, it makes faithful non-SMC emulation harder because the emulator must track more potential modification sites as the unroll depth grows.

6 Dynamic SMC tamper-proofing design primitives

Dynamic generation generalizes the same ideas while removing many of the practical limits of handwritten assembly. Figure 4 shows a representative AsmJit fragment. Important features include the ability to bind labels at arbitrary offsets, compute code sizes for fragments before final emission, and emit RIP-relative x86-64 addressing directly. These features are especially valuable because x86-64 relative encodings can change size depending on the final distance to a label, making precise layout difficult to calculate by hand.

Our implementation used three code blocks: a prologue, the checksum body, and an epilogue.

```
if (lbl) cb.bind(*lbl);
cb.adc(x86::rbx, x86::ptr(x86::rcx));
cb.sets(x86::dl);
cb.setp(x86::al);
cb.shl(x86::al, 2);
cb.or_(x86::dl, x86::al);
cb.shl(x86::dl, 3);
cb.xor_(destlbl ? x86::byte_ptr(*destlbl, 1) :
    x86::byte_ptr(x86::rip, PAGE_SIZE), x86::dl);
// add 1 to skip the prefix byte
cb.lea(x86::rcx, x86::ptr(x86::rcx, 8));
```

Figure 4: Example snippet of dynamic SMC generated via AsmJit.

```
Manufacturer: GenuineIntel
DisplayModel: 0x9E DisplayFamily: 0x6
Architecture Performance Monitoring Version: 4 Number of Performance Counters per Logical Processor: 4 PMC bit
      width: 48
The page size for this system is 4096 bytes.
Code segment base: 0x7FF7B4521000 size: 0x8200 End address: 0x7FF7B4529200
Number of pages: 2 Unrolled loop size per page: 148 Intro bytes: 0x5F Code size: 0x1B
```

Figure 5: Representative processor, operating-system, and dynamic-layout details used during experimentation.

The prologue and checksum body were first emitted with placeholder addresses to determine exact code size. Then the prologue and the required number of checksum copies were arranged to fill two pages of memory, after which the epilogue was appended and the whole region emitted into RWX pages. The prologue and epilogue preserve registers and satisfy calling conventions.

In practice, a dual-page layout with each unrolled checksum unit writing to the corresponding offset in the alternate page proved sufficient and close to ideal. We also used SMC to implement branch-free termination so that each unrolled iteration avoids conditional control flow. The final termination patch can still induce an occasional pipeline clear, but this is infrequent enough to resemble a branch misprediction rather than a persistent cost.

When executing the design, we logged processor and layout details to make the experimental configuration explicit. Figure 5 shows a typical output block. Most of these values are discovered through `CPUID`, page size queries, or the layout computations already described.

7 Performance results

For the basic static SMC model, the measured execution time is on average approximately 7.9× slower than the non-SMC variant shown in Figure 3; the measurements are given in Table 4. Among the clocks tested, `RDTSCP` is the most informative and the most stable for fine-grained timing. Each timed interval also contained one console write and one performance-counter driver call, which add the same kind of overhead to every variant. The minimum, average, and maximum values reveal heavy-tailed noise caused by infrequent context switches, but the central mass of the samples remains tight enough to support practical thresholding.

This distinction matters when interpreting the tables. The maximum values should not be read as the “typical” execution time of the checksum; they are dominated by rare scheduler interference and other system-level disturbances. The minima, by contrast, are much more stable and better represent the uncontended baseline. For deployment, this suggests calibrating thresholds from empirical quantiles or repeated-sample windows rather than from raw maxima alone.

Timer	SMC			Non-SMC		
	Avg.	Min.	Max.	Avg.	Min.	Max.
RDTSC	8305286	7466370	17053914	1052981	959776	11039730
RDTSCP	8305389	7466412	17054428	1053077	959834	11040460
Interrupt Time	32044	28806	65800	4064	3704	42610
Perf. Counter	32045	28807	65802	4064	3703	42617
Precise UTC	32045	28806	65803	4065	3704	42622
Tick Count	3	0	16	0	0	16
Multimedia	3	1	7	0	0	4

Table 4: 10,000 runs on a hexa-core Intel Core i7-9750H with 16 GB RAM under Windows 10; the checksum covered the entire 32.5 KiB (0x8200-byte) code segment.

Timer	SMC			Non-SMC		
	Avg.	Min.	Max.	Avg.	Min.	Max.
RDTSC	467241	414132	2157922	42300670	39599882	74294904
RDTSCP	467286	414188	2158056	42300840	39600010	74295022
Interrupt Time	1803	1598	8328	163202	152779	286641
Perf. Counter	1803	1599	8327	163204	152779	286644
Precise UTC	1803	1598	8328	163206	152779	286647
Tick Count	0	0	16	16	0	32
Multimedia	0	0	2	16	15	29

Table 5: Dynamic SMC measurements under the same environment and conditions as Table 4.

The dynamically generated SMC variant is substantially more competitive. As shown in Table 5, it is approximately 90.5× faster than the semantics-faithful, timing-constrained non-SMC variant and about 17.8× faster than the basic static SMC implementation (Table 4). This version used two pages and wrote to the corresponding offset in the opposite page; using three or four pages did not materially improve results.

The spread in the dynamic measurements also gives a rough sense of how often the operating system perturbs the benchmark. Using the `RDTSCP` figures as a back-of-the-envelope estimate, the SMC variant yields

$$\frac{10{,}000 \cdot (467{,}286 - 414{,}188)}{2{,}158{,}056 - 414{,}188} \approx 304, \qquad \frac{304}{10{,}000} \approx 3.0\%,$$

which suggests that only a small fraction of runs suffer a substantial task-switch style disturbance. Performing the same estimate for the semantics-faithful non-SMC variant gives roughly 7.8% (Table 5, non-SMC columns, with the minimum subtracted in the same way). The exact percentages should not be over-interpreted, but they reinforce the practical point: the dominant source of variance is scheduler noise, not instability in the SMC design itself.

We can also inspect SMC-induced pipeline clears directly. Table 6 reports the aggregate counts across 10,000 runs. The C variants are affected by context switching and unrelated system activity, which can include just-in-time execution from other processes. The naive SMC variant, unsurprisingly, triggers a large number of clears because the checksum enters its own introspective region and continues modifying code there. The unrolled static and dynamic loop variants remain the most instructive result: they still incur some clears at the start of each unrolled iteration, showing that the current termination logic leaves room for improvement, but the reduction relative to the naive SMC case is dramatic.

Variant	Event count
C	4514
SMC	83198361
C Loop	20986
SMC Loop	4492871
Dyn SMC Loop	4597855

Table 6: SMC pipeline-clear occurrences over 10,000 runs.

8 Efficiency analysis

Any non-SMC reproduction of the mechanism must generally spend more space-time resource than its SMC counterpart. Even when both are asymptotically linear, the relevant practical quantity is the constant factor: the goal is to engineer a large enough constant c such that $c_{\text{non-SMC}}\, n \gg c_{\text{SMC}}\, n$. A stronger checksum family might even force super-linear behavior in a faithful non-SMC emulation, though we do not rely on that here.

Precisely proving optimality is difficult because the cost is dominated by microarchitectural details. Searching for an optimal non-SMC emulation strategy is, in general, combinatorial. At the same time, handwritten assembly is not automatically optimal: compilers can sometimes discover improvements that are hard to find manually. This makes high-level prototyping for code generation a valid engineering strategy.

Several architectural lessons are important. First, modifying instructions that are effectively in flight can reduce performance to something reminiscent of a late-1990s Pentium III-class core, because pipeline-clearing behavior destroys the benefits of deep speculation and out-of-order execution. Second, branching is expensive even with a branch predictor, so non-SMC emulations that require many conditionals are intrinsically disadvantaged. Third, an ISA-independent design rule is that SMC should conservatively avoid modifying the currently executing page. Finally, timing scores are stable enough in practice for detection, with the largest perturbations caused by operating-system task switches; those can be mitigated, though not eliminated, by adjusting process or thread priority.

9 Using primitives as part of a tamper-proofing scheme

The model requires a timer $T(t)$ tailored for reliability at the tth query and a checksum function $C(c)$ over code or data c. The tamper-proofing predicate $P(T, C)$ returns true for the unmodified protected code (when its timing lies within the calibrated bounds) and false for every modification that changes the checksum state or violates the time bound. In practice, security comes from deploying many checksum units so they cannot all be removed or overridden easily. A state machine with precomputed validation states can implement $P(T, C)$ for a controlled subset of states, while a reset function restores the SMC to a canonical state and bounds the state-space growth.

The timer-validity thresholds can be estimated empirically on representative hardware or upper-bounded conservatively. The most useful policy parameters are the allowed CPU-usage range in the operating-system environment, the length of the protected computation, and the number of consecutive failed timing scores that trigger fatal termination or recovery. In a practical deployment, these thresholds are tuned together with code layout rather than in isolation.

10 Security assumptions and threat model

We state the adversary, its goal, the assumptions the scheme needs, and what is proved. The statements are formalised and checked in Lean (Section 12); platform behaviour enters only as explicit contract fields (timing bounds, which capabilities a privilege level has), not as proved facts about hardware.

Adversary levels. L0: patches the protected bytes before execution. L1: user mode; may write to the protected process while it runs, single-step or breakpoint it, run it under emulation, and hook operating-system timer interfaces. L2: kernel or hypervisor; additionally may give instruction fetch and data reads different views of memory and may trap or offset the time-stamp counter. L4: additionally has physical writers such as DMA devices. (L3 is reserved and has the capabilities of L2 in the model.) The *goal* is that a protected byte other than a legitimately varying SMC opcode byte is executed in modified form and no alarm is effective at any point of the (unbounded) run. An alarm is effective when a checksum unit with at least one response byte, all unmodified, sees a wrong result, or when $k \geq 1$ consecutive checks of one unit exceed its time threshold τ. The SMC opcode bytes are reset by every check, so tampering with them is outside the goal.

Assumptions. A1 (coherence, replaces "von Neumann architecture"): the fetch and data views are one image. A2 (coverage, replaces "cannot neutralize all functions"): every protected byte lies in the region of some checksum unit, each unit's own code is protected, the coverage graph of the units (a unit covers another when its region contains the other's code) is strongly connected, and at least one unit is never modified. A3 (timing gap, replaces the non-existence of an equally fast non-SMC implementation): the native check time is at most τ, and τ is below the lower bounds on emulation, on a debug trap, and on mirroring an opcode write. A4 (no physical writer): no device other than the protected thread writes the protected memory; a larger-area defense narrows the window between check and use but cannot close it. A5 (time integrity): every timer reading equals the elapsed time. A6 (persistence): $k \geq 1$, and a modification persists through the next k scheduled checks of each unit that covers it. A7 (response coverage): each unit has at least one response byte, and its response bytes are part of its own protected code and are not SMC opcode bytes. A8 (non-compensation): a modification inside the region of a running unit changes the unit's result. The checksum is deliberately not collision resistant, so A8 is an assumption about the design, and it cannot be dropped: two modified quadwords can restore the final sum and opcode state.

Results. (T1) Under A1–A8 and the platform contract, no adversary achieves the goal: if a protected non-opcode byte is ever executed in modified form, an effective alarm occurs somewhere in the run, not necessarily before that execution. (T2) With separate fetch and data views, a run that follows the SMC kernel's opcode trajectory needs one mirroring event per opcode-changing iteration (except the last), and under A3 and A5 a mediated check admitted by the cost contract that contains such an iteration is a timing failure. This does not say that every split-view attack is detected. (T3) Each of the eleven items A1, A2 (coverage, unit code protected, strong connectivity, one unmodified unit), A3, A4, A5, A6, A7 and A8 is necessary within the model: for each there is a well-formed system and an adversary that satisfies the other ten and wins. (T4) With the inline time-stamp counter, every well-formed timeline at level L0 or L1 satisfies A1, A4 and A5. Conversely there is a winning adversary against a system that satisfies every assumption except one: A5 at L1 with an operating-system timer interface and at L2, A1 at L2, A4 at L4. The level capabilities are fields of the contract, not facts about real systems. With a time source outside

the adversary's control, of resolution r and jitter j, slowdowns of at least $j + 2r$ are detected by a threshold calibrated on native runs.

Scope. Not claimed: protection against code injected outside the checksummed bytes, against register tampering outside timed intervals, or against a modification that is undone before the next covering check; guarantees against L2 and above without an external clock; equivalence to cryptographic attestation. The mechanism is a local runtime-integrity technique intended to raise attacker cost on commodity systems, and the proofs cover the model above, not the hardware.

11 Summary and outlook

We have developed a tamper-proofing toolkit that binds code integrity to timing and runtime-generated SMC. The central idea is to make faithful reproduction of both semantics and execution timing costly for an adversary, even when the host environment is untrusted.

The main engineering lesson is that naive SMC is too slow, but carefully designed SMC is not. Reliable clocks such as `RDTSCP` provide stable enough timing signals for practical thresholding, while loop unrolling and cross-page self-modification largely eliminate the worst pipeline stalls. Dynamic SMC then turns these gains into a flexible deployment strategy that is materially faster than a semantics-faithful non-SMC emulation.

The approach is still bounded by platform-specific threshold tuning, operating- system scheduling noise, and the practical difficulty of measuring some hardware events without privileged support. Promising directions for future work include cross-modifying code in multithreaded settings, checksum kernels whose best non-SMC emulation incurs super-linear overhead, automated blueprint generation for dynamic SMC, and validation on additional ISAs.

Overall, polymorphic SMC paired with carefully engineered timing semantics offers a practical path to runtime integrity for trusted code executing in untrusted environments.

12 Formal verification

The threat model of Section 10 is formalised and machine-checked in Lean 4 (toolchain `leanprover/lean4:v4.35.0-rc3`, no Mathlib) [15]. The Lean development is registered with the Palomar registry as entry `PALOMAR-2026-10-18-000004`, version 1, and is published at https://github.com/GregoryMorse/SelfModTamper, release tag `palomar-v1`, commit `0a48a85e0f1884a50c6abdcf8617bd315a6b59c2`. The registered statement consists of the four results T1–T4 of Section 10, proved for the abstract model defined there (checksum units, adversary levels L0–L4, assumptions A1–A8, and a cost contract). Platform behaviour, that is, what a privilege level can do, which timing bounds hold, and what an operating system or processor provides, enters only as explicit fields of that model, not as a proved fact about real hardware or software. The same repository contains Lean checks of further statements of this paper (layout and timing arithmetic, checked as consistency of the reported numbers; the separation of Harvard and von Neumann architectures; a forgery of the checksum from three modified quadwords) that are not part of the registered statement. The repository file `lean/PAPER_MAP.md` assigns each claim of the paper to a Lean declaration, a cited platform fact, or the class of observation. The Lean development is supporting material: the paper's measurements and listings do not depend on it, and what it checks is limited to the statements described above. The listings, native tests and measurements are evidence for the implementation, not proofs. The corrections to the text of the first version that this version makes were found while formalising the statements.

AI assistance disclosure

Anthropic Claude Opus 5.5 in high thinking mode and Anthropic Claude Sonnet 5.5 in medium thinking mode, used through the Claude Code command-line agent, assisted with the Lean formalization (statement drafting, proof development and proof repair), the validation scripts, the repository documentation, the comparison of the first version with the formalization, and the drafting of the corrections made in this version. The Lean kernel checks the proofs; the author reviewed the formal statements, the corrections to the text and this manuscript, and assumes full responsibility for the final work.

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